

COMPSYS 700A - Literature Review

Assessing the feasibility of an FPGA-based doppler ground speed sensor
using affordable COTS components

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1 Introduction

High-performance vehicles require precise, real-time chassis velocity data relative to the ground. In Formula Student events, where maximizing longitudinal acceleration during the standing-start drag event is the primary objective, the vehicle control system must continuously balance motor torque against the tyre slip ratio. As shown in Figure 1, the relationship between slip ratio and tractive force is non-linear. This means that maximum grip is achieved at a specific slip target, typically 10 to 15 percent, where the rubber deforms and mechanically interlocks with the track surface. Below this window, the tyre is under-driven, while above it the tyre enters an unstable free-spinning region and forward acceleration collapses.

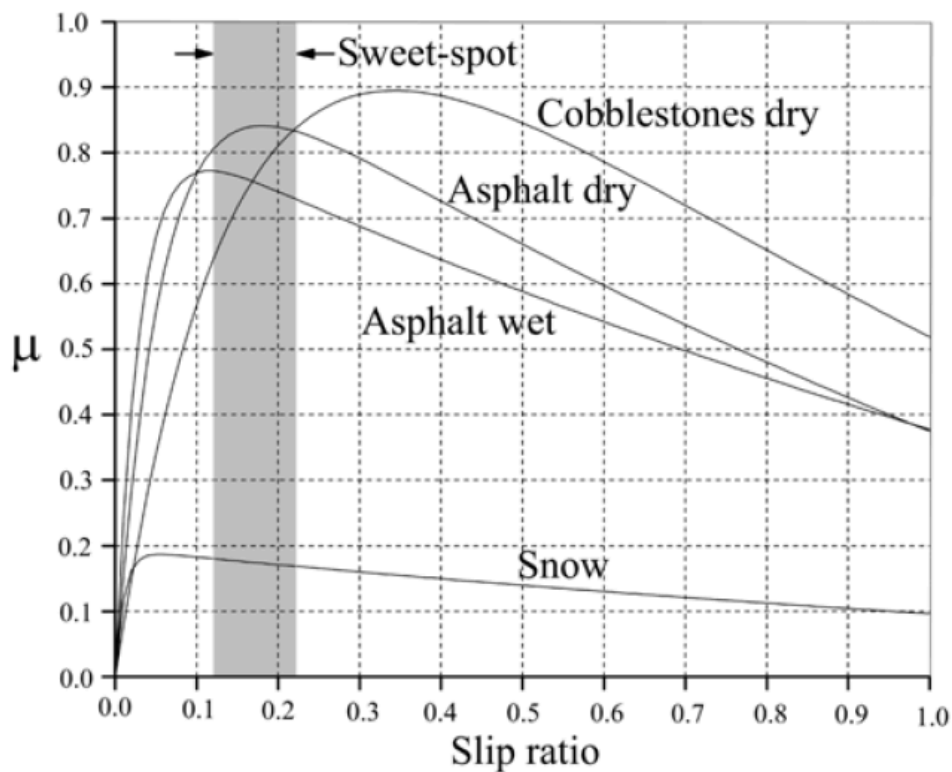


Figure 1: Burckhardt tyre model showing adhesion coefficient (μ) versus slip ratio for common racing surfaces. Maximum tractive force occurs in the 10–15% slip window.

Electric vehicles typically infer speed from rotary encoders on the motors. During a hard launch, however, wheel speed and chassis speed diverge. A control system relying solely on encoder data cannot distinguish the optimal grip region from traction-breaking wheel spin, and cannot therefore close the traction control loop around the correct slip target. An independent Ground Speed Sensor (GSS) prevents this ambiguity by measuring true chassis velocity directly, enabling true slip ratio computation and therefore real-time closed-loop traction control.

The practical issue is cost and reliability. Professional GSS units from manufacturers such as Kistler or Sensoric Solutions cost thousands of dollars a month for rentals which is unrealistic for a student team operating on a limited seasonal budget. Optical flow sensors have been explored academically as a lower-cost option, but they depend on consistent surface texture, stable ride height, and controlled lighting, none of which can be guaranteed on a live race surface. Doppler radar addresses every one of these failure modes with microwave signals penetrating dust and

rain, passing through opaque bodywork, and measuring velocity directly from a frequency shift rather than inferred motion.

This project investigates whether a ground speed sensor built from affordable consumer off-the-shelf (COTS) radar hardware, processed on a budget FPGA, can meet the strict latency and accuracy requirements of real-time Formula Student traction control.

2 Objectives

- Verify velocity measurement accuracy using a known and controlled reference speed. A controlled constant speed testing rig will be used to determine if the COTS sensor can achieve less than 5 percent error compared to theoretical Doppler calculations.
- Quantify total system latency from analog input to serial output. This will validate if the parallel FPGA architecture can process signals within the strict 20ms deadline required to prevent traction control loop instability.
- Evaluate the limits of the sensor, with signal-to-noise ratio being assessed across varying ride heights (100mm to 300mm) and different scattering surfaces like gravel, concrete, and rubber.

3 Scope

This project validates the proposed sensor system initially within a controlled laboratory environment. Accuracy and latency testing will be conducted using a constant speed test rig as a reference, allowing repeatable, measurable conditions independent of a vehicle. Deployment on a car is explicitly outside the current scope. If bench validation meets the targets above, integrating the sensor with the University of Auckland Formula SAE vehicle and the existing MoTeC M150 data system is a natural next step, but is treated as future work rather than a project deliverable. Environmental variables unrelated to the core sensor pipeline, such as tyre thermal degradation, battery state of charge, or lateral dynamics are excluded to keep validation focused on the DSP chain from analog radar signal to serial velocity output.

4 Literature Review

4.1 The Physics of Traction and State Estimation

4.1.1 Slip Ratios and Tractive Force

To accelerate a vehicle, the motor controller applies torque to the driven wheels, increasing their angular velocity. When the angular velocity of the tyre exceeds the linear velocity of the vehicle chassis, the tyre slips against the road surface. This relative difference is quantified as the slip ratio [1].

The relationship between slip ratio and tractive force is highly non-linear. According to the Burckhardt tyre model, the adhesion coefficient between the tyre and the road increases roughly linearly with slip ratio until it reaches a specific peak [2]. For most racing tyres it is agreed that this peak falls between 10 and 15 percent slip [1]. If motor torque pushes the slip ratio beyond this point, the adhesion coefficient drops and the tyre enters an unstable sliding region where the wheel free-spins and forward acceleration drops [2].

4.1.2 The Limitations of Kinematic State Estimation

Controlling the slip ratio requires accurate knowledge of both the wheels' angular velocity and true chassis linear speed. Angular velocity is straightforward to measure with motor encoders, but obtaining chassis speed without a dedicated ground sensor requires kinematic estimation.

A common approach combines wheel speed with a longitudinal accelerometer [3]. The problem is that accelerometers measure acceleration, which includes gravity. During a hard launch the chassis pitches upward, tilting the sensor relative to the ground and allowing a gravity component to bias the longitudinal axis [4]. Velocity is computed by integrating acceleration, so even a small constant bias compounds over time and the estimated speed quickly diverges from true speed [5]. Kalman filters can fuse accelerometer and wheel speed data to correct this drift [3], but they rely on periods of zero wheel slip to recalibrate the accelerometer bias [5]. During a drag launch the wheels are deliberately kept in continuous slip, which removes exactly this recalibration opportunity and causes the filter to fail. A dedicated GSS bypasses the problem entirely by measuring ground speed directly.

4.2 Comparative Analysis of GSS Technologies

4.2.1 GNSS and Inertial Navigation Systems

Global Navigation Satellite Systems measure velocity by tracking satellite signals, providing chassis speed entirely independent of wheel encoders. Standard GPS units typically update at 1 to 20 Hz [6], introducing a phase delay that becomes increasingly detrimental during rapid acceleration. GPS also suffers signal dropout near large trackside structures, making it unreliable near grandstands or barriers common at Formula Student venues. Keskin et al. measured a strong correlation between GPS velocity error and vehicle acceleration across a range of receivers. The error magnitude was found to increase significantly during rapid speed changes, which is a condition a traction control system must handle [6].

4.2.2 Optical Flow Sensors

Optical non-contact sensors track microscopic surface texture using high-speed cameras. Eriksen et al. built an optical GSS for a Formula Student car using camera mouse displacement sensors and achieved errors below 3.5 percent at 100 Hz [7], but the implementation exposed funda-

mental reliability problems. The sensor requires a stable focal length and consistent lighting [7]. On a race car, suspension travel, chassis roll, and pitch continuously change the sensor-to-ground distance. Smooth concrete, rubber debris, or standing water all eliminate the surface texture that the algorithm depends on, causing large velocity errors. These conditions are not uncommon on a racing surface.

4.2.3 Lidar vs. Radar Technologies

Lidar and Radar both sense the environment by emitting energy and measuring the return, but operate on fundamentally different wavelengths. Lidar uses near-infrared light, which is heavily attenuated by water vapour, fog, and tyre smoke [8], and requires an exposed optical lens that becomes unusable when coated in track debris. Standard automotive Lidar infers velocity by differentiating range measurements, which amplifies noise [8].

Radar operates in the millimetre-wave band. These wavelengths pass through dust, rain, and plastic enclosures [8], allowing the sensor to be fully sealed. More importantly, radar directly measures the Doppler frequency shift of the return, meaning velocity is a direct output, not a derivative. Albrecht et al. demonstrated a 122 GHz radar GSS achieving 100 Hz update rates with accuracy matching a professional Kistler optical reference sensor at even lower noise outputs [9].

4.3 Radar Signal Processing

4.3.1 Principles of Millimeter-Wave Doppler Radar

Automotive radars operate in the mm-wave bands, typically 24 GHz or 77 GHz, where wavelengths range from 1 mm to 10 mm [10]. The Doppler frequency shift is $f_d = 2\frac{v}{\lambda} \cos(\theta)$, where v is velocity, λ is the wavelength, and θ is the angle between the radar beam and the direction of motion [11]. At 24 GHz the sensitivity is approximately 160 Hz per m/s, meaning even small velocity changes produce easily measurable frequency shifts.

For ground speed measurement, distance to the road is irrelevant. Pulsed and Frequency Modulated Continuous Wave (FMCW) architectures carry significant added complexity allowing for extra features such as range, which is unreasonable dead weight for a sensor that only needs a single velocity output [10]. A Continuous Wave (CW) radar transmits an unmodulated sine wave and dedicates all transmit power and processing bandwidth to the Doppler shift. Placentino et al. showed a 24 GHz CW receiver achieving under 3 percent velocity error [11], which is sufficient to keep a Formula SAE tyre within its optimal slip window of 10-15% [1].

The $\cos(\theta)$ term introduces a systematic error when the chassis pitches. If the vehicle nose rises during launch, θ increases and the sensor undershoots true speed. A practical mitigation is the Janus configuration, where there are two beams aimed to the front and the back of the chassis at equal but opposite inclined angles [9]. The cosine error in each beam has opposite sign, so averaging their outputs substantially reduces the pitch-induced bias [9]. This approach is standard in aircraft Doppler navigation and is directly applicable to a racing launch where sustained pitch is the primary issue.

4.3.2 Spectral Estimation and Peak Tracking Techniques

The core signal processing task is extracting the Doppler peak frequency from a noisy spectrum. The simplest approach, time-domain zero-crossing, counts how often the received signal crosses zero volts to estimate frequency [12]. This method fails under two main conditions present

during racing, low-frequency flicker noise where the signal baseline shifts and produces false crossings, and specular reflection off wet track surfaces where the return amplitude drops below the noise floor entirely, at which point zero-crossing produces no processable output [13].

FFT based methods allow the Doppler data to return the velocity as a spectral peak rather than a threshold crossing. Before any peak estimator is applied, Constant False Alarm Rate (CFAR) thresholding can be used to suppress noise in the spectrum. Rather than applying a fixed amplitude cutoff, CFAR estimates the local noise floor from the frequency bins surrounding each possible peak bin and sets the detection threshold adaptively, so genuine relative peaks in noisy regions are retained while isolated noise spikes are rejected [10]. This pre-processing step improves the reliability of whatever peak estimator follows.

These FFT based frequency domain methods are robust to both failure modes previously seen in zero-crossing. Shariff et al. compared two approaches to extracting a single velocity value from the spread Doppler spectrum [13]. The simpler of the two is the Center-of-Mass Algorithm (CMA), which computes the power weighted average frequency of all spectral bins above a noise threshold. It runs approximately 30 times faster than the alternative method covered next. [13]. The weakness is that any bins above the threshold contribute to the average, so noise spikes or clutter returns pull the result away from the true peak. At signal-to-noise ratios above 20 dB both methods perform well, but at 10 dB the CMA becomes unreliable while still technically producing an output [13].

The Cross-Correlation Algorithm (XCA) addresses this by exploiting the known shape of the return signal. Because a radar antenna illuminates a cone of ground rather than a point, each scatterer within the beam contributes a slightly different Doppler shift, and the combined power spectrum approximates a Gaussian envelope [13]. XCA generates a reference Gaussian characterized by the antenna beam width and slides it across the spectrum measured, taking the position of maximum correlation as the Doppler peak. Due to it matching the entire spectral shape rather than individual bin amplitudes, isolated noise spikes do not shift the result.

Shariff et al. validated this difference between the two methods experimentally. On a water covered surface where return power dropped by approximately 10 dB, CMA produced an average error of 29.2% and returned repeated zero-speed failure flags, while XCA maintained 8.0% error under the same conditions. On dry asphalt both algorithms achieved 0.5–1.5% relative error [13]. Given that wet and contaminated surfaces are among the test conditions for this project, XCA is the preferred algorithm, with CMA retained as a computationally cheaper fallback if FPGA resource constraints within a reasonable budget make XCA impractical.

4.3.3 Spectral Leakage and Windowing Functions

FFT processing operates on finite time blocks and assumes the block repeats periodically, so any mismatch between the waveform at the boundaries creates a discontinuity. In the frequency domain a discontinuity requires infinite harmonics to represent, meaning the signal energy is smeared across all bins, which is called spectral leakage [14]. The resulting side lobes raise the noise floor around the Doppler peak and reduce the accuracy of peak location.

To suppress leakage, the time block is multiplied by a window function that tapers smoothly toward zero at both edges before the FFT is applied [14]. This eliminates the boundary discontinuity at the cost of effectively reducing the contribution of samples near the edges. Davidovici et al. compared several window functions and found that the Hamming window is

preferable to the commonly used Hanning window for this application. The Hanning window tapers completely to zero, discarding edge information entirely, while the Hamming window tapers to a small non-zero value of 0.08 that cancels the largest adjacent side lobe to the main peak we are looking for [14]. To recover the edge information lost by any window, successive blocks are overlapped by 50 percent using Welch’s method, which also smooths the spectral estimate without adding latency.

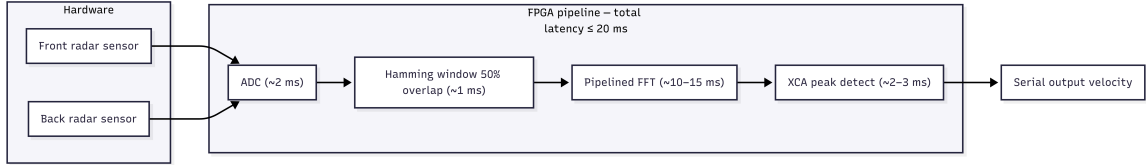


Figure 2: Summary of the signal processing pipeline from radar hardware to velocity output.

4.4 Hardware Limitations and FPGA Acceleration

4.4.1 The Microcontroller Bottleneck

Standard microcontrollers execute instructions sequentially, meaning they cannot ingest new ADC samples while simultaneously computing a transform on the previous buffer. Misans and Terauds noted this constraint directly, observing that zero-crossing was chosen in part because FFT-based methods were considered too computationally demanding for their target microcontroller platform [12]. Benchmarking by Heo et al. confirmed that a memory-based software FFT at the resolution required for accurate Doppler peak extraction exceeds practical real-time deadlines on sequential processors, motivating the move to dedicated hardware [15].

4.4.2 FPGA Architectures for DSP

FPGAs implement DSP as dedicated parallel hardware logic rather than a sequential instruction stream. For the FFT, a pipelined architecture accepts new samples every clock cycle and produces output with a fixed latency, rather than a memory-based architecture that stalls while computing [15]. Heo et al. compared pipelined and memory-based FFT architectures directly and found the pipelined design reduced processing latency by over an order of magnitude at the cost of significantly higher logic resource consumption [15], making it the appropriate choice where a hard timing deadline applies.

Alongside architecture, the choice of number format determines both signal fidelity and resource consumption. Fixed-point integer arithmetic is efficient but introduces quantization noise that elevates the spectral noise floor around the Doppler peak, degrading peak location accuracy. Heo et al. showed that 16-bit half-precision floating-point arithmetic reduces this quantization noise to negligible levels, but requires 1.69 times the hardware resources of an equivalent integer implementation [15]. Folded DSP multiplier reuse can reduce FPGA resource consumption [16] at the cost of added pipeline stages. This is a tradeoff that must be considered when balancing latency with cost.

4.4.3 The Research Gap

No published work addresses the intersection of all three constraints specific to this project. High-performance radar GSS systems such as Albrecht et al.’s 122 GHz sensor [9] demonstrate excellent accuracy but use custom integrated transceivers and dedicated hardware well outside a student budget. ADAS radar literature [10] focuses on multi-target FMCW processing with

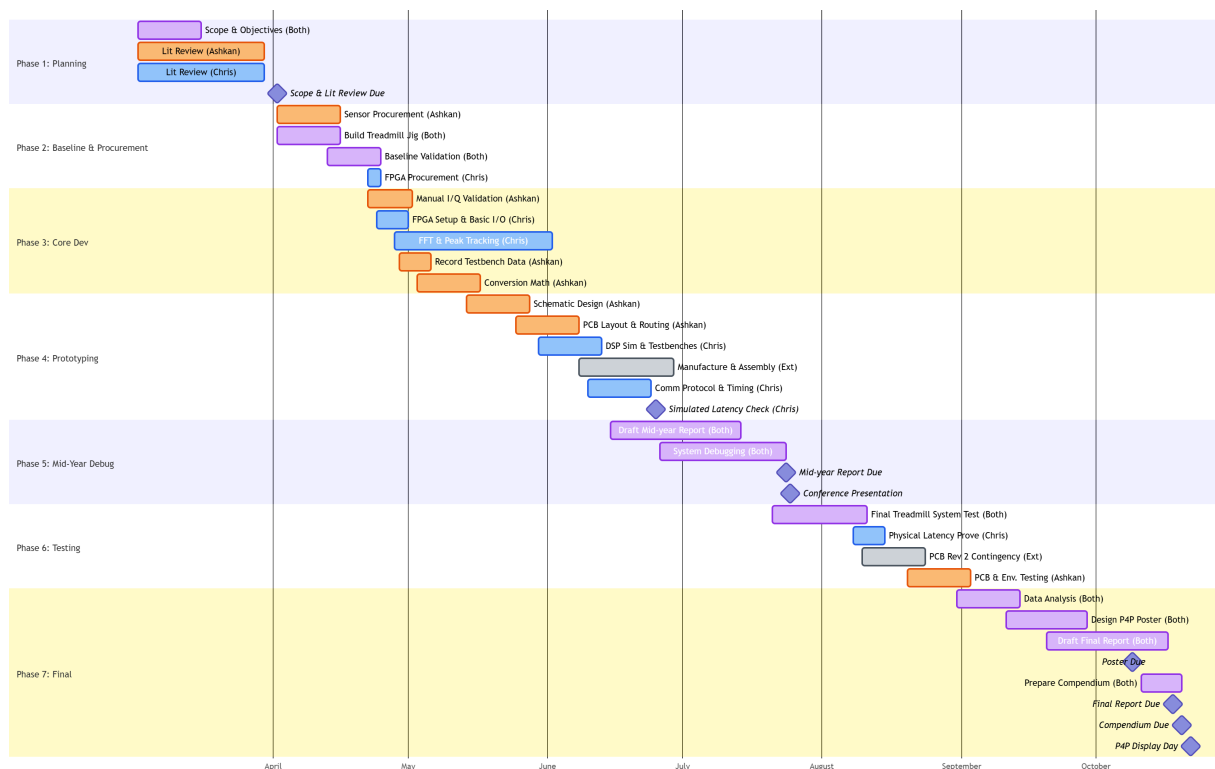
latency tolerance measured in hundreds of milliseconds. Traffic enforcement radar [11], [13] operates from a fixed roadside post where latency is irrelevant. FPGA DSP research [15], [16] addresses general signal processing architectures without the motorsport specific constraint of fitting a robust floating-point pipeline onto an affordable COTS FPGA that must also meet the latency deadline.

The gap is then that no published work demonstrates a CW Doppler GSS pipeline running on an affordable COTS FPGA with verified end-to-end latency under 20 ms in a motorsport context. A Formula Student team building a car needs a sensor that costs under a few hundred dollars in components, outputs velocity with under 5 percent error, and delivers that output within 20 ms of the physical event. The result of addressing this gap determines whether high performance traction control is accessible to student motorsport teams without commercial sensor budgets.

5 Acknowledgement of AI usage

Claude and Gemini were used during as writing and research tools during the literature review. They were used for document formatting, explaining technical concepts, and improving report structure and wording.

6 Project Timeline



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